

Lesson 0009 — Project walkthrough and interview practice

~70–90 MIN

Synthesis

ANCHORED TO: [README.MD](#) · [DOCS/ARCHITECTURE.MD](#) · [EVALUATION/M5_LIVE/REPORT.MD](#)

The final lesson is a rehearsal: explain the complete data path, cite the evidence, defend the trade-offs, and state the current limitation and next reversible experiment.

HOW TO USE THIS LESSON

Read the explanation, inspect the visual, make a prediction, and then compare it with the repository evidence. The goal is a model of this project that you can explain—not a list of framework slogans.

PREREQUISITES AND SCOPE

This is the synthesis lesson. Complete Lessons 0001–0008 first, or follow the links in each section when a term is unfamiliar. You are not expected to memorize line numbers; you are expected to distinguish contracts, evidence, inference, and uncertainty.

LEARNING OBJECTIVES

By the end, you should be able to explain the system at three levels of detail, trace one frame across every boundary, distinguish a model result from a system result, attach conditions and denominators to measurements, defend a trade-off, state an unknown without guessing, and propose a test whose outcome would change your conclusion.

Project walkthrough and interview practice

EXPLAIN THE SYSTEM · SHOW THE EVIDENCE · OWN THE LIMITS

READING ORDER

1 → 2 → 3 → ... → 12

1 90-SECOND SUMMARY

State the problem, approach, inputs, outputs, and current M5 status without overclaiming.

2 INPUT PATH



Vendor RGB and depth images enter a ROS 2 node through named topics.

3 SYNC + CONVERT



ApproximateTimeSynchronizer pairs frames; cv_bridge turns messages into arrays.

4 MODEL PATH



Letterbox, TensorRT YOLOv5 inference, decode, confidence filtering, and NMS produce candidates.

5 GEOMETRY GATE



Depth statistics decide whether a candidate looks like a raised cube-like object.

6 OUTPUT CONTRACTS



The node publishes standard Detection2DArray, vendor ObjectInfo, and a debug image.

7 ARTIFACT CHAIN



best.pt → best.onnx → best.engine separates learning, graph exchange, and deployment.

8 EVIDENCE TIMELINE



M2 trained, M3 exported, M4 ran on Jetson, M4c filtered, M5 exposed the room gap.

9 TRADE-OFFS



YOLOv5s + TensorRT matches the vendor target; geometry adds evidence without replacing the model.

10 FAILURE ANSWER

Say what failed, what was measured, what was ruled out, and what remains uncertain.

11 NEXT STEP



Fine-tune on the room domain, re-export, repeat regression, then decide whether to promote.

12 INTERVIEW LOOP



Explain, point to a file, cite a measurement, name a limitation, and propose the next test.

PROJECT ANCHOR

Anchors: README.md · docs/architecture.md · models/README.md · evaluation/m5_live/report.md

EXPLANATORY MAP · NOT A MEASUREMENT

Orientation map for the final walkthrough. The detailed sections provide the evidence, uncertainty, and answer rubric. Click the PNG to open it at full resolution; the source SVG remains available for inspection.

Retrieval warm-up

Answer from memory before revealing the cues. Difficulty here is useful: it strengthens retrieval and shows what deserves review.

1. Why is 100% per-class frame-hit coverage not 100% accuracy?

► Reveal recall cues

2. State the artifact chain and one verification question at each boundary.

► Reveal recall cues

3. What does M5 PARTIAL mean in one evidence-backed sentence?

► Reveal recall cues

The question this lesson answers

Can you explain the complete colored-cube system, its trade-offs, its failures, and its next step as an engineer? This final lesson is a walkthrough and a rehearsal. The strongest answer is not “the model works”; it is a short, evidence-backed chain from camera input to current limitation.

INTERVIEW RULE

Separate **implemented**, **measured**, **planned**, and **unknown**. That vocabulary makes the explanation trustworthy.

A RELIABLE ANSWER SHAPE

Use four moves: **claim** what you think is true; name the **evidence** and its scope; state the remaining **uncertainty**; propose the smallest **next test** that could distinguish competing explanations. “I do not know yet” is a strong engineering answer when followed by an observable test.

Word	What it permits you to say
Implemented	The code or configuration exists; it may still be untested.
Verified	A named check passed under named conditions.

Word	What it permits you to say
Measured	A number was observed with a stated unit, denominator, artifact, and dataset or bag.
Inferred	The evidence supports an explanation, but alternatives remain.
Planned	A future action is documented; no outcome exists yet.
Unknown	The current evidence cannot decide; name the next discriminating observation.

1. The 90-second project summary

“This project detects `red_cube` , `green_cube` , and `blue_cube` from a HiWonder JetRover camera. A ROS 2 node synchronizes RGB and depth, preprocesses the RGB frame for a YOLOv5s TensorRT FP16 engine, decodes and filters candidates, checks depth geometry, and publishes standard and vendor-compatible outputs plus a debug image. The model artifact chain is `best.pt` → `best.onnx` → `best.engine` . M5 is PARTIAL: the node and empty-scene rejection run, but the current model does not reliably propose the room cubes at confidence 0.50. The next planned intervention is positive-domain fine-tuning followed by the same regression gates.”

This summary is intentionally compact. The following sections give you the evidence needed when an interviewer asks “how?” or “what failed?”

USE PROGRESSIVE DISCLOSURE

Start with a one-sentence purpose, continue with the 90-second architecture and status, and expand only where the listener asks. A strong explanation is not the longest possible explanation; it is the shortest one that preserves the important boundaries and can be expanded without contradiction.

2. Whiteboard the runtime path



Use this order when explaining the system. It keeps data, computation, and contracts visible.

Trace one concrete frame, not just component names: a timestamped RGB message and nearby depth message are paired; RGB pixels are converted and letterboxed; the engine emits a dense tensor; confidence filtering and NMS reduce it to boxes; padding is undone; depth statistics may reject a box; then the surviving data is adapted into ROS messages. At each arrow ask, “What representation crosses this boundary, and what information could be lost or changed?”

MODEL VERSUS SYSTEM

The detector is one subsystem. A model can produce a good box while synchronization, coordinate reversal, depth filtering, or publication later removes or corrupts it. Conversely, a healthy ROS message rate proves the pipeline is alive, not that the boxes are correct. System evaluation must localize both kinds of failure.

3. Name the real input and output contracts

Role	Current contract
RGB input	/depth_cam/rgb/image_raw , sensor_msgs/Image
Depth input	/depth_cam/depth/image_raw , sensor_msgs/Image
Camera info	/depth_cam/rgb/camera_info , intrinsics update
Standard output	/cube_detections , vision_msgs/Detection2DArray

Role	Current contract
Vendor output	<code>/cube_detections/vendor_objects , interfaces/ObjectsInfo</code>
Debug output	<code>/cube_detections/debug_image , annotated sensor_msgs/Image</code>

Class-ID caution: the current model and node use 0=blue_cube , 1=green_cube , 2=red_cube , but docs/project-definition.md still documents another order. Call out and reconcile this before interpreting a class index or training again.

A contract includes more than a type name. For images it also includes width, height, encoding, timestamp, frame identity, and coordinate convention; for tensors it includes shape, axis meaning, dtype, units, and preprocessing; for detections it includes class mapping, score meaning, box coordinates, and whether an empty message is valid. Type-compatible data can still be semantically wrong.

4. Describe what the model does—and what it does not do

The node's `_letterbox()` prepares a square model input. `_infer()` runs the TensorRT engine and calls `_decode_yolov5_output()` . The current export emits `[1, 7, 8400]` : one image, seven fields, and 8,400 candidate positions. The decoder transposes it into candidate rows, takes the best of three class scores, removes rows below the confidence threshold, converts center/size boxes to corners, reverses padding, and applies per-class NMS. This produces a much smaller set of detections in the original RGB frame.

The model does not itself publish ROS messages, estimate floor depth, or prove that a confidence score is true. Those responsibilities belong to the node and geometry/output stages.

5. Explain the depth gate precisely

`compute_geometry()` estimates a local reference depth from a surrounding annulus, applies the literal `depth > floor + threshold` selection, estimates that subset's extent using intrinsics, and measures aspect ratio and depth spread. `decide()` returns a verdict and reason. Missing or sparse per-box depth is kept with a low-quality reason; sufficient statistics can reject as flat, aspect, or non-planar. The filter is a post-filter, not a replacement detector, and its current selection sign is physically reversed for z-depth, the distance measured along the camera's optical axis.

Say both the intention and the implementation: the intended idea is that a cube top is raised relative to nearby floor, but in camera z-depth a nearer raised surface normally has a *smaller* value. The current greater-than predicate selects farther pixels instead. This is a confirmed semantic defect in the present code, so historical KEEP/REJECT counts verify reproducibility of that code path—not physical correctness.

6. Explain the artifact chain

best.pt PyTorch/Ultralytics checkpoint used for learning and export.	best.onnx Static exchange graph checked and exercised with ONNX Runtime.	best.engine TensorRT FP16 artifact built for the Jetson runtime.
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A change at the first boundary requires verification at every later boundary. The engine is not just a renamed checkpoint.

7. Tell the evidence timeline

Stage	Evidence-backed statement
M2	<code>best.pt</code> trained; validation mAP@0.5 0.954 on 9 images/21 instances.
M3	<code>best.onnx</code> exported and checked; output shape <code>[1, 7, 8400]</code> .
M4/M4b	TensorRT engine loaded; each predicted class ID appeared in every frame of the saved 30-frame camera set. This is frame-hit coverage, not ground-truth accuracy.
M4c1	Current filter removed the named distractor candidates; real-blue preservation was partial; the physical predicate is not validated.
M5	Live node ran; empty scene kept 0; cubes at conf 0.50 kept 0; verdict PARTIAL.

Notice the scope attached to every number. “0.954” is not universal accuracy: it is mAP@0.5 on a small named validation split. “Every frame” is not object recall without labels and one-to-one matching. “0 keeps” on an empty bag is encouraging for that scene but does not establish a general false-positive rate. Measurements become trustworthy explanations only when their artifact, inputs, conditions, metric, and denominator travel with them.

8. Defend the main trade-offs

- **YOLOv5u-style model + TensorRT:** the checkpoint uses the anchor-free Ultralytics head while the project keeps “YOLOv5s” as shorthand; ONNX is the portable boundary and the engine is Jetson-oriented.
- **Depth geometry:** is intended to add a shape/elevation check for candidates, but its current $\text{depth} > \text{floor} + \text{threshold}$ predicate has the wrong sign for z-depth and must not be presented as validated height reasoning.
- **Approximate sync:** accepts nearby RGB/depth timestamps using a 50 ms slop rather than demanding exact equality.
- **Vendor boundary:** consumes vendor topics/contracts but does not modify vendor packages or the boot service.
- **Read-only evidence:** keeps evaluation separate from irreversible promotion or physical actuation.

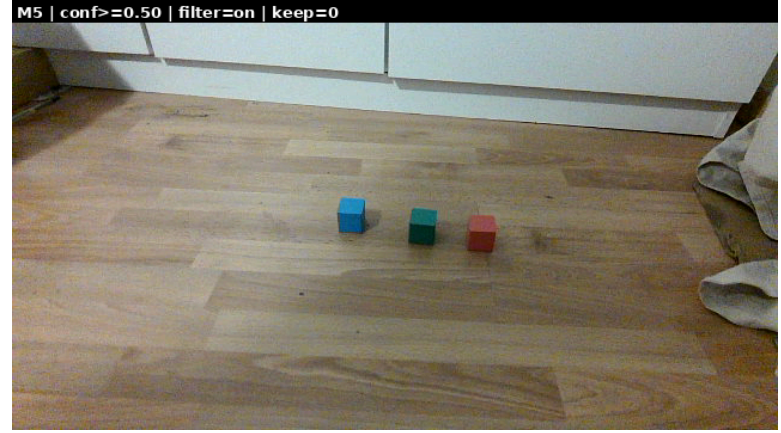
A trade-off answer should name both benefits and costs. For example, a static 640×640 engine makes deployment predictable and efficient, but fixes the input contract and may lose small-object detail compared with a larger input. Approximate synchronization increases the chance of receiving pairs, but moving objects can shift between the two timestamps. A lower confidence threshold can recover weak cubes, but also admits more false candidates. Defending a choice means showing why its cost is acceptable for the measured operating conditions—not claiming that it has no cost.

9. Answer the failure question

“Why does it fail?” A disciplined answer is: validation and earlier saved-camera results show the artifacts can run, while M5's live room distribution differs. At confidence 0.50, the sticker-off cubes bag produced 0 kept detections across 460 output messages; at 0.25 only two green keeps appeared across 439 messages. Candidate logs make a positive-detection/domain-gap hypothesis plausible, and geometry cannot create a missing candidate. However, these counts are not ground-truth accuracy, the geometry predicate has a confirmed depth-sign defect, and the evidence does not prove the model is the only cause or rule out every ROS/preprocessing issue. The responsible next step is a controlled experiment across those boundaries. [M5 evidence · artifact evidence]



Observed input: the cubes are visible to a human.



Observed pipeline output: the debug HUD reports `keep=0` .

DO NOT JUMP FROM CORRELATION TO CAUSE

The live room differs from the training/validation data and the detector misses cubes there, so domain shift is a strong hypothesis. That pattern alone does not prove causation. A decisive investigation compares raw engine outputs across controlled preprocessing and scenes, then counts what each downstream filter removes. Keep alternative hypotheses alive until the observation distinguishes them.

10. Practice follow-up questions

Interactive: reveal the shape of a strong answer

Start with the data contract: topic names, message types, and the synchronized RGB/depth pair.

The “80-frame” plan itself contains conflicting totals and must be reconciled before collection, as Lesson 0008 explains. In an interview, saying this does not weaken the project; it demonstrates that you can distinguish a proposed direction from an executable experimental specification.

SELF-CHECK RUBRIC

A strong answer names the input/output contract, traces the stages in order, separates general concepts from this artifact's numbers, gives at least one measured result with its test conditions, names one limitation without exaggeration, and ends with a reversible test. A weak answer relies on “AI magic,” treats confidence as truth, calls keep counts accuracy, or presents a plan as completed evidence.

11. Check your understanding

Common shortcut	More accurate statement
“8400 detections”	8400 candidate positions before confidence, NMS, and optional geometry filtering.
“0.90 means 90% correct”	A learned score whose real reliability must be measured on representative labelled data.
“Validation was 95.4% accurate”	mAP@0.5 was 0.954 on a named 9-image validation split.
“TensorRT improved the model”	TensorRT changed execution; behavior and speed require separate verification.
“Depth detects cubes”	Depth post-filters boxes that YOLO already proposed.

What makes the 90-second summary strong?

- 1. ☐ It connects system flow, evidence, limitations, and next actions clearly.
- 2. ☐ It lists implementation details while leaving measured limitations unstated entirely.
- 3. ☐ It reports successful outcomes while omitting evidence and unknowns entirely.

What is the current engineering verdict?

- 1. ☐ M5 PASS: runtime and positive room detection both meet targets.
- 2. ☐ M5 FAIL: the node and TensorRT engine have never run.
- 3. ☐ M5 PARTIAL: runtime works, but positive room detection misses targets.

What is the responsible next step?

- 1. ☐ Modify vendor packages, then evaluate whether detections begin appearing reliably.
- 2. ☐ Capture representative room positives, fine-tune, then rerun every gate.
- 3. ☐ Promote the lowest-loss checkpoint, then lower all runtime thresholds immediately.

Which statement is the most defensible?

1. ☐ The detector is 95.4% accurate everywhere.
2. ☐ The ROS node publishes, so cube detection works.
3. ☐ The checkpoint recorded $mAP@0.5=0.954$ on its 9-image, 21-instance validation split, while M5 room detection remained partial.

12. Tiny exercise: record your own walkthrough

1. Set a timer for 90 seconds.
2. Explain the system using the pipeline diagram and the project summary, without reading the prose.
3. Answer “what failed?” with at least two exact M5 measurements.
4. Answer “what next?” with the planned intervention and the rollback boundary.
5. Then give a five-minute version that traces one frame and labels every major statement as implemented, measured, inferred, planned, or unknown.

Observable result: a recording or written transcript that contains the inputs, artifact chain, geometry role, outputs, M5 PARTIAL verdict, and next experiment. Use the rubric below to self-check.

Data

Can you name the topics and message types?

Evidence

Can you cite counts without turning them into guesses?

Boundaries

Can you say what is planned or still unknown?

13. Explain it back

Final explain-back: “Walk a technically skeptical interviewer from a JetRover image to the current M5 verdict. Explain where the model, ROS 2, TensorRT, depth geometry, and evaluation each contribute—and where each cannot help.”

Finish with one reversible next step.

What this lesson intentionally postpones

This lesson intentionally postpones robot manipulation, navigation, architecture swaps, detailed neural-network mathematics, the actual fine-tuning run, and repair of the geometry predicate. Those are not missing implied successes: they are separate implementation and experiment tasks. The goal here is an honest systems explanation grounded in the current repository.

Primary references and next lesson

- Ultralytics YOLOv5 documentation — model-family and YOLOv5u architecture context.
- ROS REP-118: Depth Images — the depth convention needed to explain the geometry defect honestly.
- COCO detection evaluation — why keep counts without ground-truth matching are not accuracy.
- Project README and the linked canonical documentation.
- Curriculum plan — scope, teaching contract, and project truth.
- Return to Lesson 0001 for a foundations refresher, or review the full curriculum plan.

CONTINUE WITH THE TEACHING AGENT

Ask the agent to clarify anything uncertain, challenge your explanation with a counterexample, or assess your explain-back. A learning record should be created only after you demonstrate the idea in your own words.

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Course home

Quick reference

Mission